Which buildings waste energy?

Not which use the most, but which use more than they should, given what they are and where they are.

Buildings are among the largest contributors to energy use and carbon emissions.

NTU Learn to Hack: MATLAB Edition 2026 · Predicting Building Energy Consumption · built in MATLAB

github.com/JoshiiWahWah/ntu-matlab-hackathon-2026

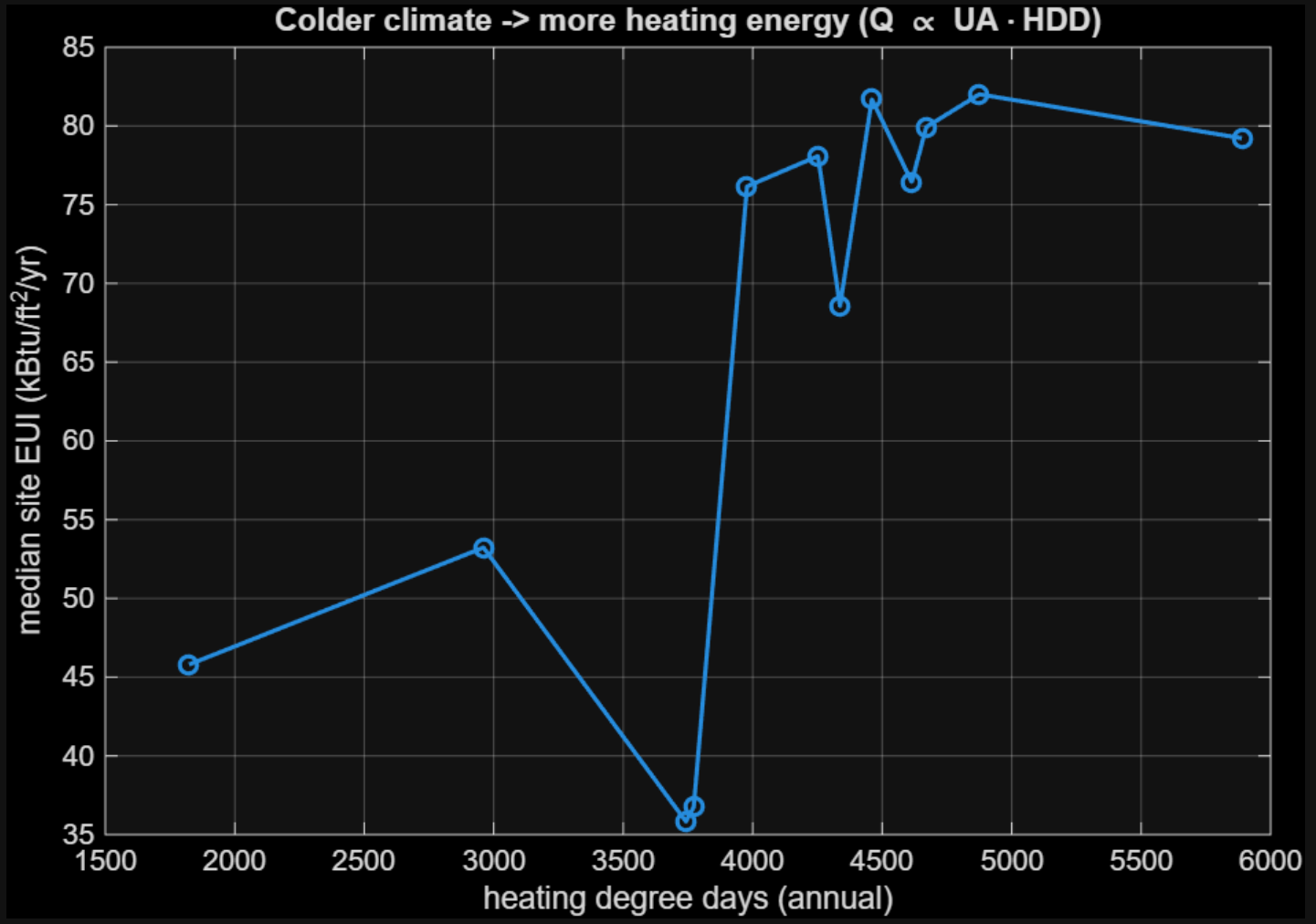
Energy balance: Q ≈ UA · ΔT summed over the year

Heating demand scales with heating degree days.

Features we engineered:

degree days · seasonal mean temperatures · annual temperature range · log floor area · building age

Median EUI: ≈ 36–53 in milder climates vs ≈ 68–82 in colder ones (kBtu/ft²/yr)



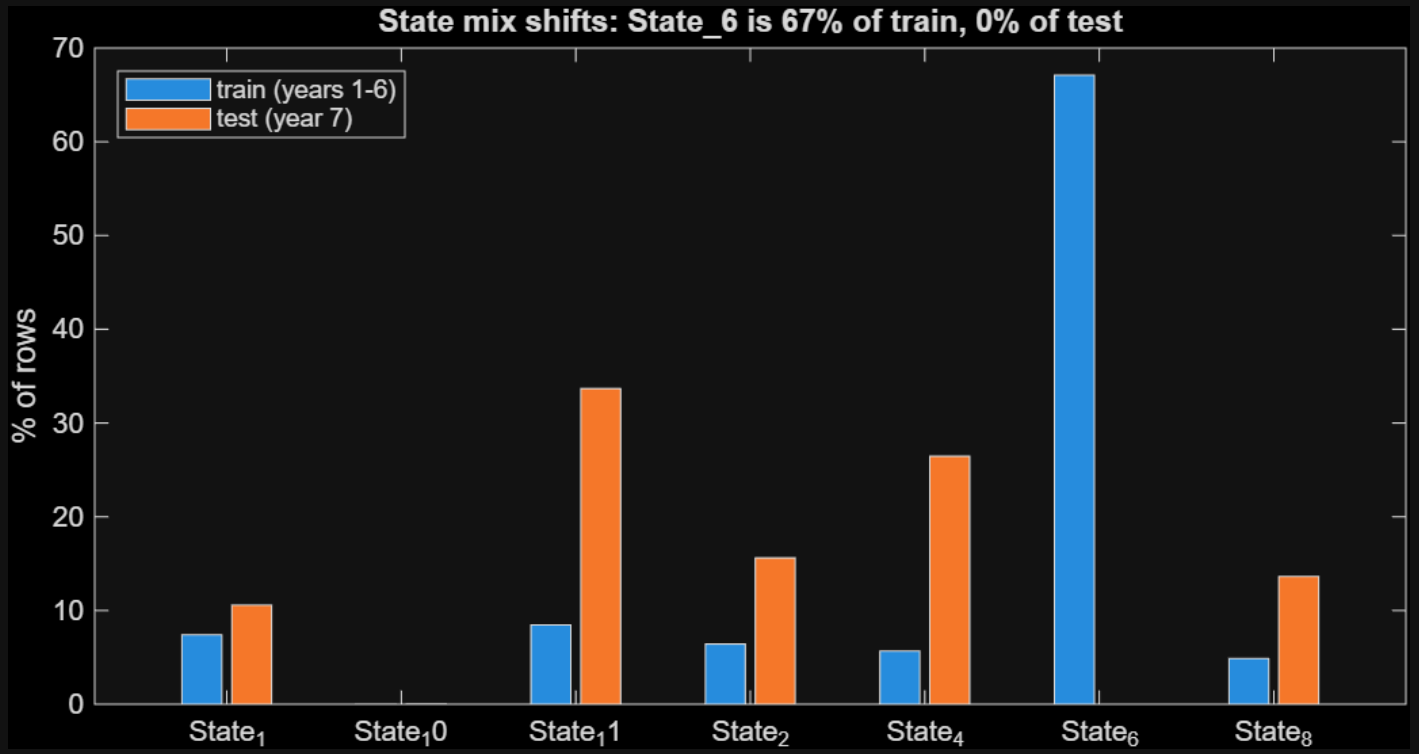
Finding 1: the test set comes from different states

State_6

67.1% of train

0% of test

→ we also report a test-like score on non-State_6 buildings



Finding 2: missingness shifts between train and test

Wind and fog columns, share missing

56.0%

train

89.8%

test

A model would learn a pattern that doesn't exist at test time → dropped

Finding 3: the same buildings recur, but without an id

63,907

buildings reconstructed from static attributes

state · class · facility type · floor area · year built · elevation

Past energy use becomes a feature

Computed only from earlier years: no leakage, proven by a unit test

22.1% of test rows have past energy data

HONEST VALIDATION

Validate the way the task works

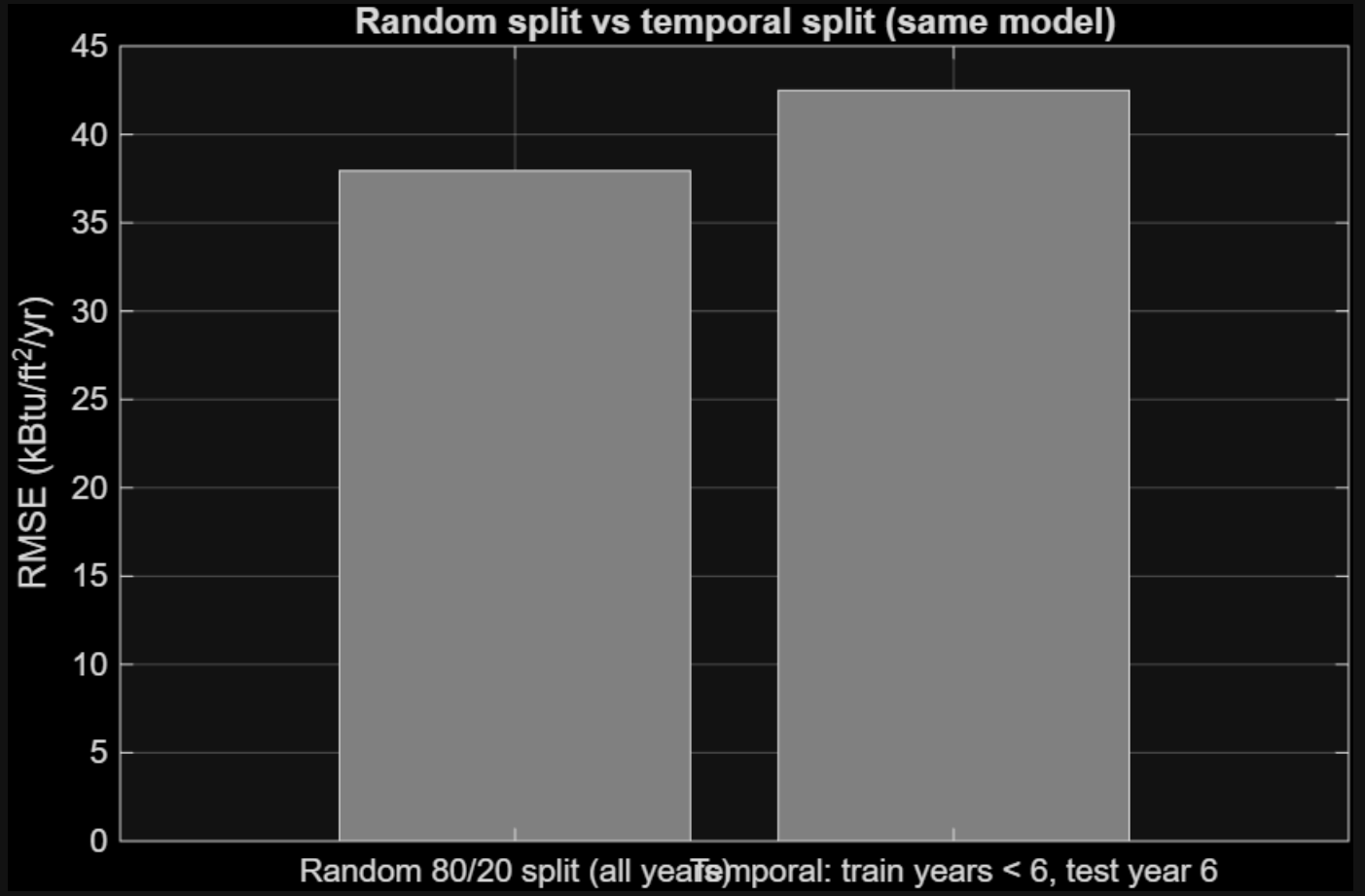
Train years 1–5 → predict year 6, never seen

Same model, same features:

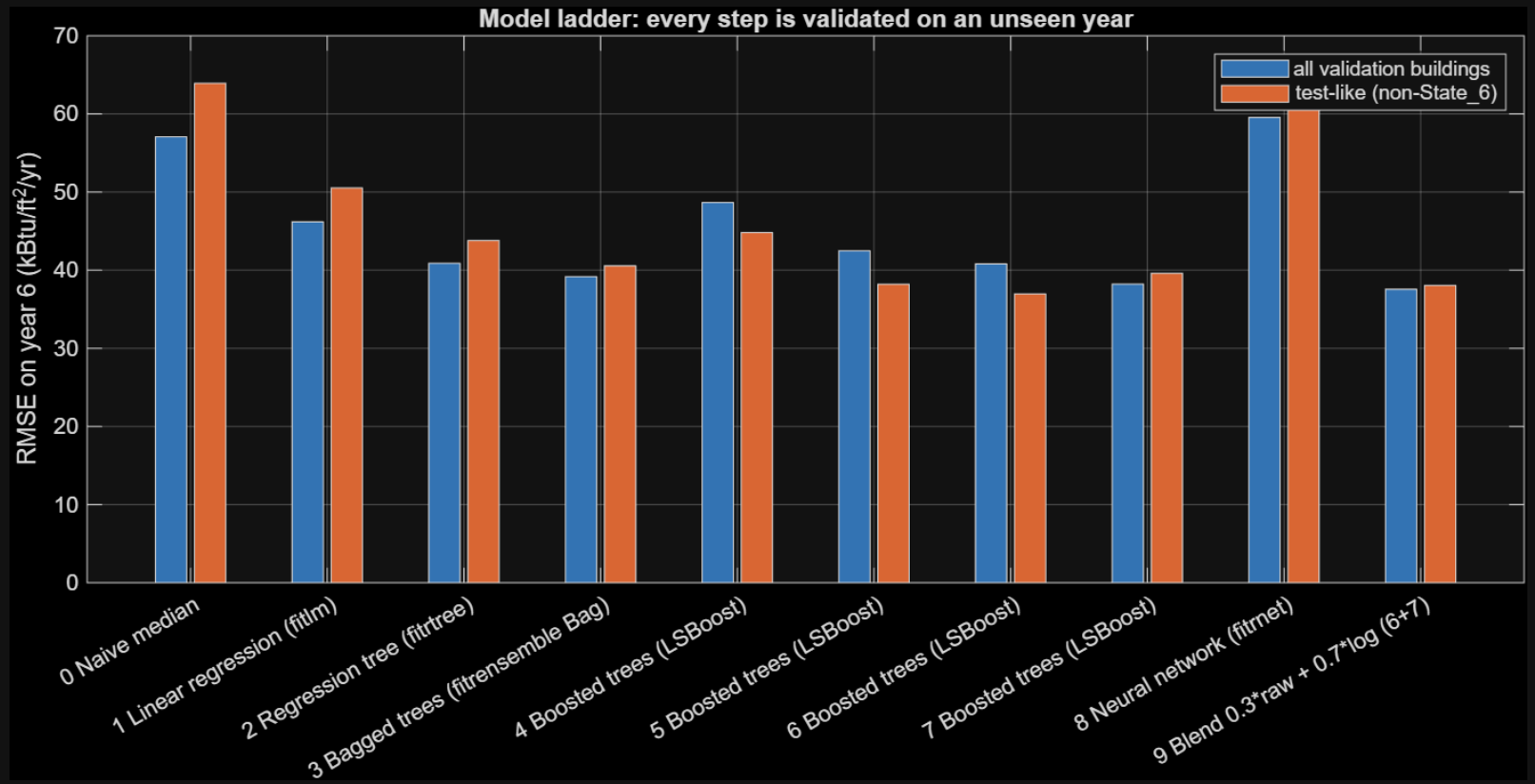
Random split 37.9

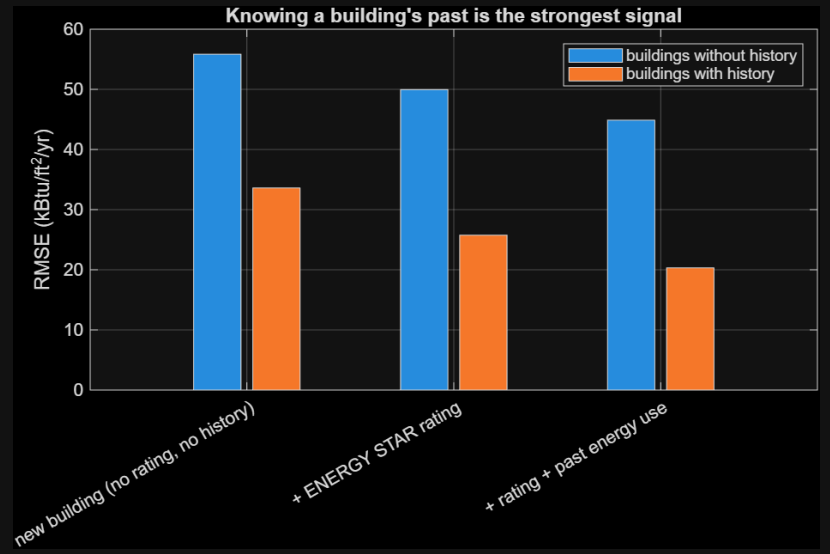
Temporal split 42.5

We report the honest temporal number.



A ladder of 10 models, each answering one question





Best: raw + log boosted-tree blend

RMSE 37.6

−34% vs naive · R^2 0.57

Linear 46.2 → single tree 40.9 · buildings with history: RMSE 20

A MATLAB-specific lesson: missing values in trees

Without surrogate splits, a row missing the split variable stops at that node.

Our first run barely learned from buildings without a rating.

Fix: encode missing values explicitly (src/treeInputs.m)

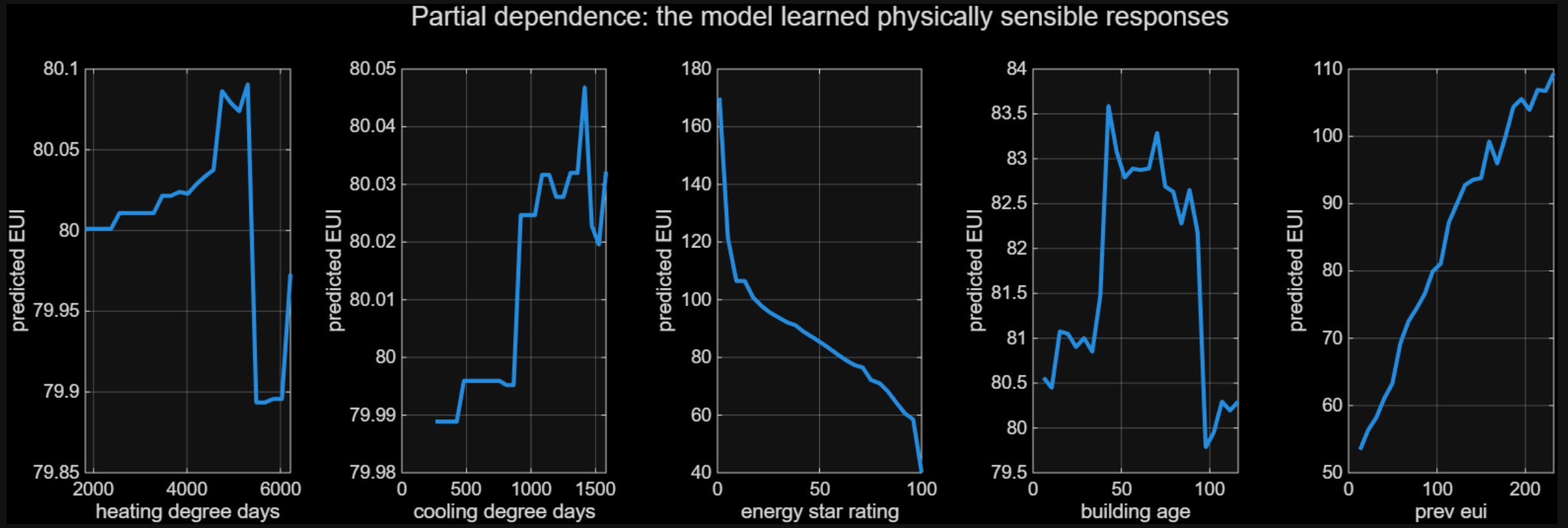
blend RMSE

46.8 → 37.6

first full run → after the fix

INTERPRETABILITY

What the model learned



Higher ENERGY STAR score → lower energy use · past use predicts future use

Honest caveat: weather is recorded per state and year, so once the state is known the degree-day features add little.

IMPACT

From prediction to action: the retrofit shortlist

Cold-start model = what a typical building of the same type, size, age and climate should use

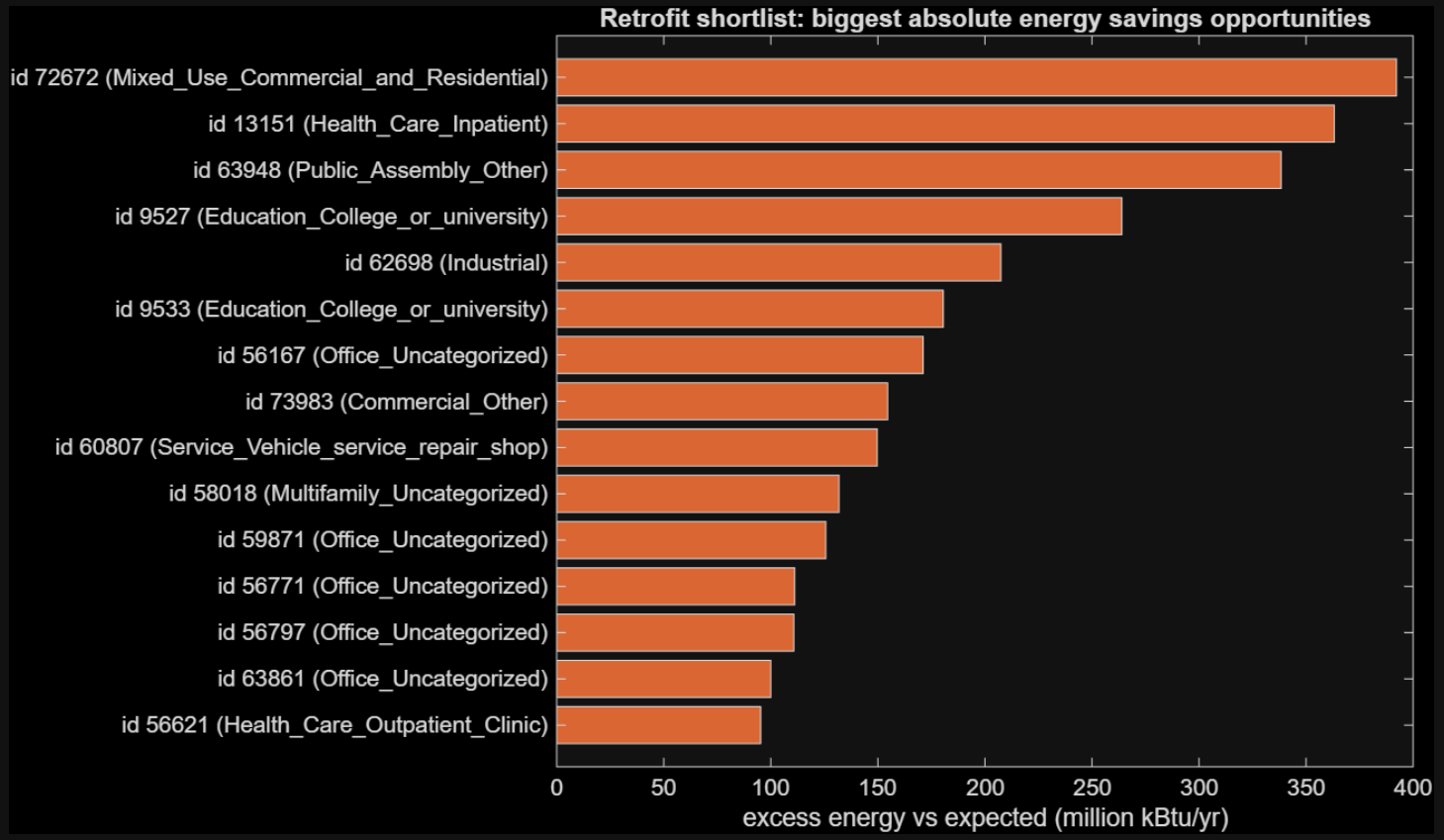
1,123 buildings

5% of the stock

1.55×10^{10} kBtu/yr above expected

= 5.3% of all energy used

Ranked by absolute excess energy



One command, reproducible, tested

>> main

Raw CSV → submission and figures, in one command

Modular functions with argument validation

Unit tests in matlab.unittest · validated submission file

Every number in our README was printed by MATLAB

NEXT STEPS

Monthly meter data

Occupancy

Optimising which retrofits to fund first

Thank you

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